Handling Vector Data

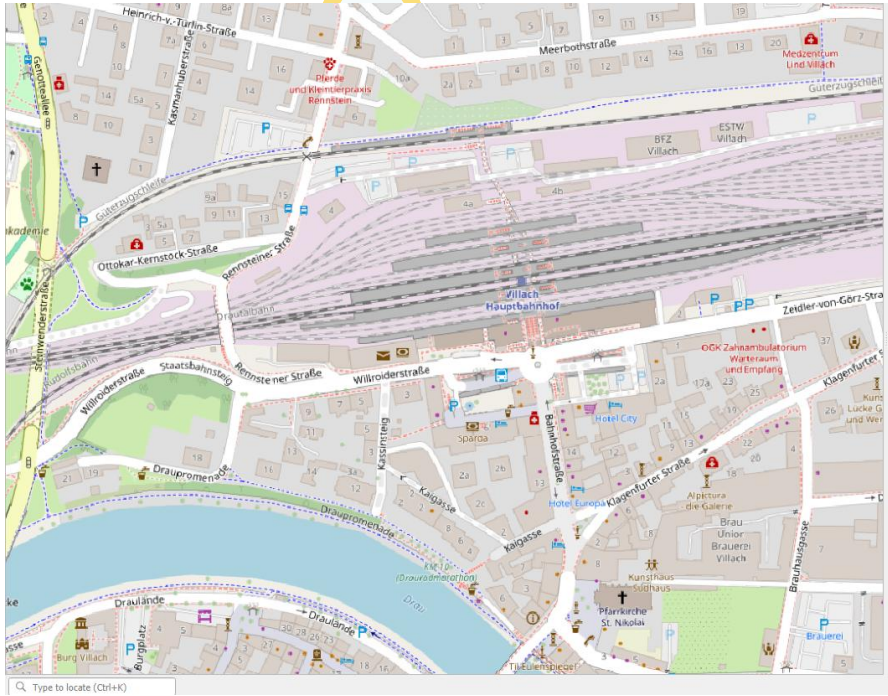
Lab_04: Network Analysis

402.467 / Applied GIS Fundamentals

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04 November 2026

Refresh our memories



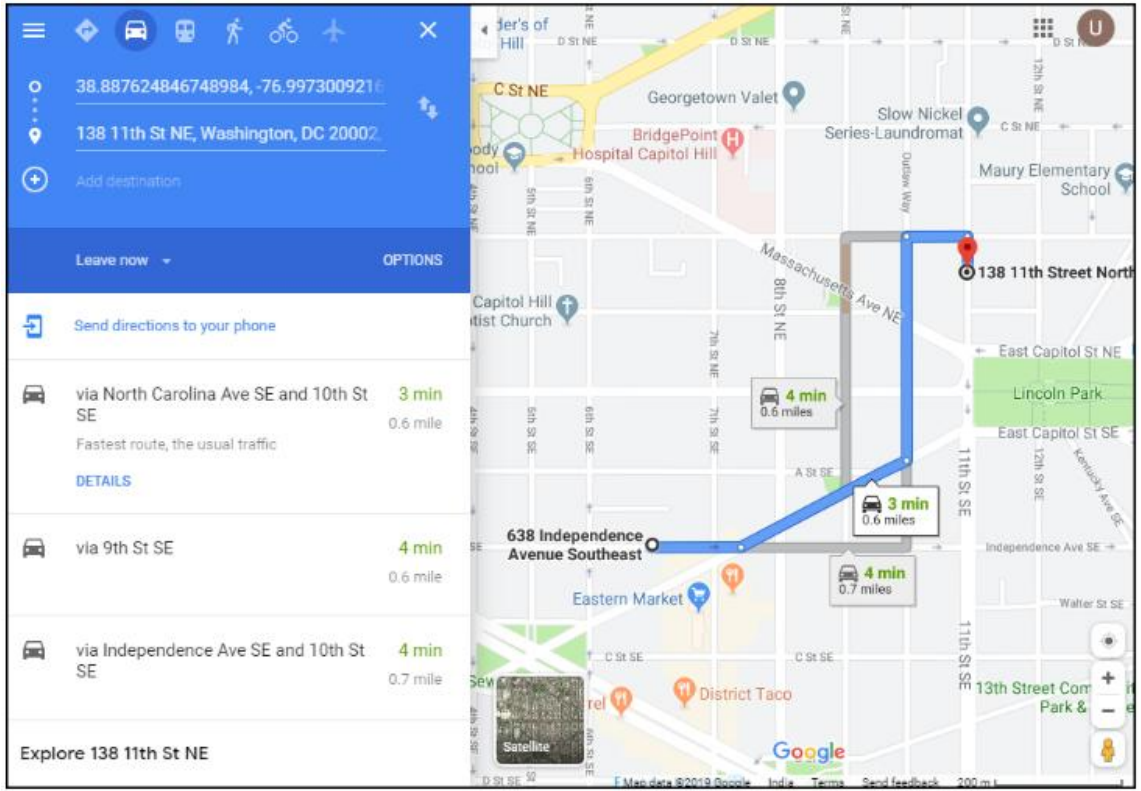


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Network analysis

-Q: How does Google Maps find the fastest route?

-A: It uses network analysis. A set of GIS tools that model and analyze movement along connected lines!



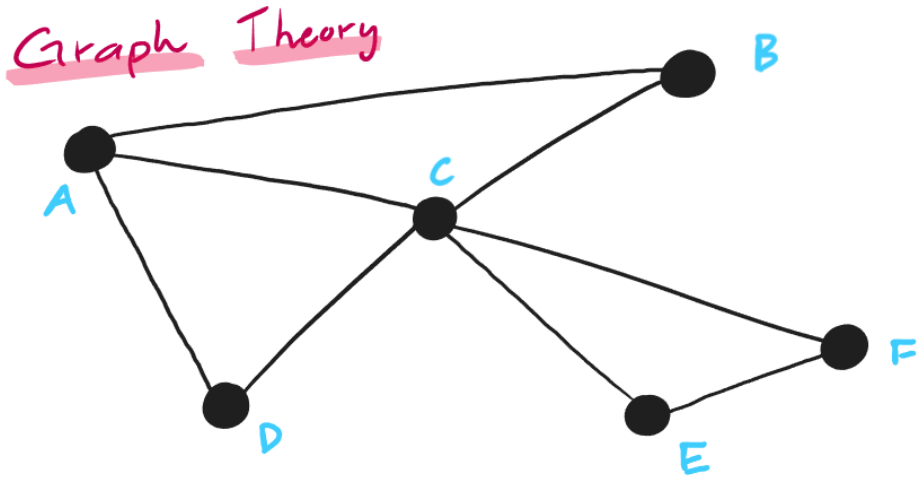
@Qgistutorials

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Graph theory

A graph (also called "network") is defined as a set of points, which are interconnected by a set of lines

A graph is a mathematical structure to model relationships between objects. It represents relationships between objects in a structured way

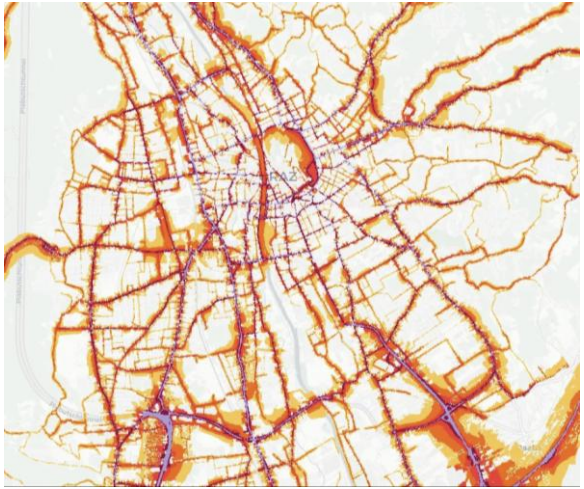


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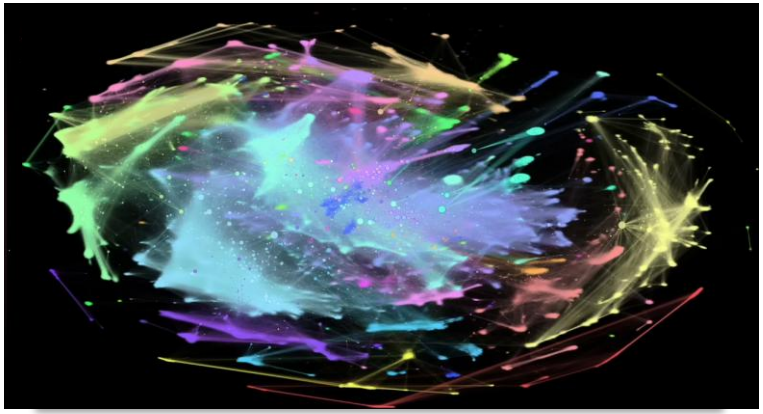
Examples

How loud is Graz?



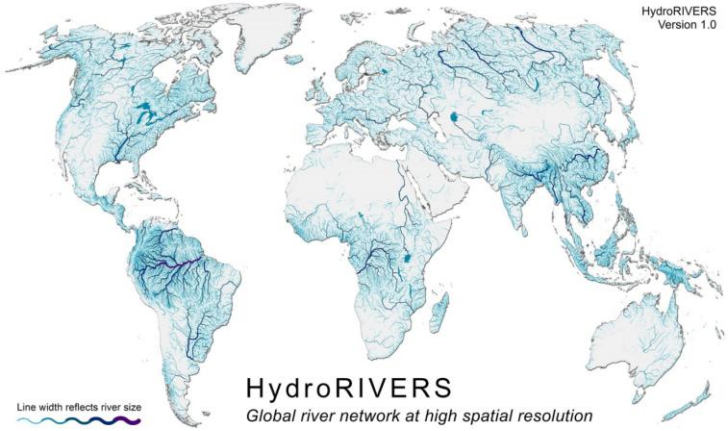
@Decibel Shield

Map of Wikipedia, clustered by digital communities



@Aadumb

River network



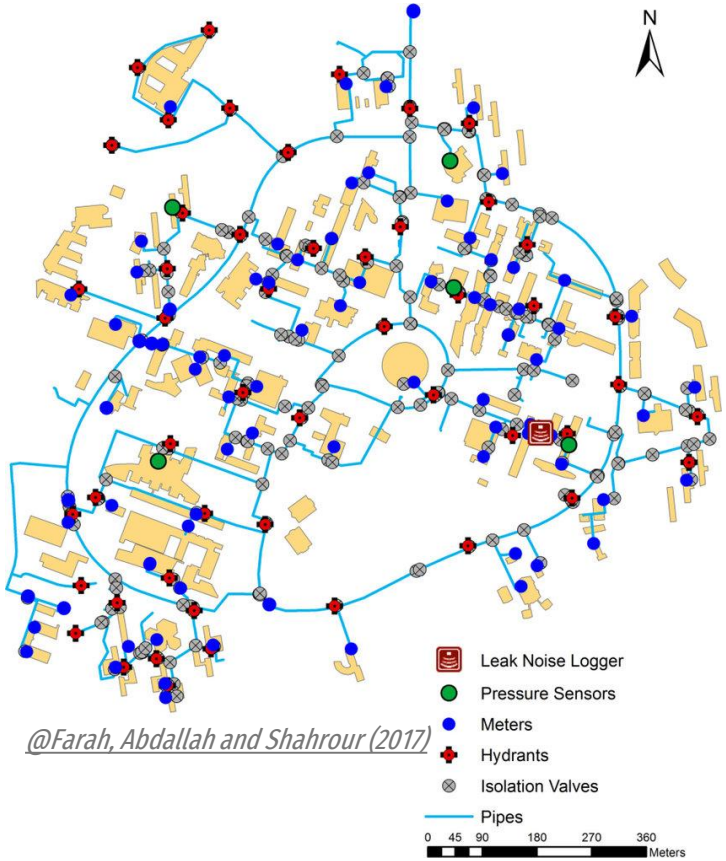
@Hydrosheds

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More examples

🚰 Smart water network at Lille 1 University

- Water network management
- Detection of leakage or abnormal use
- Map pipes, valves and sensors in GIS
- Monitor water consumption



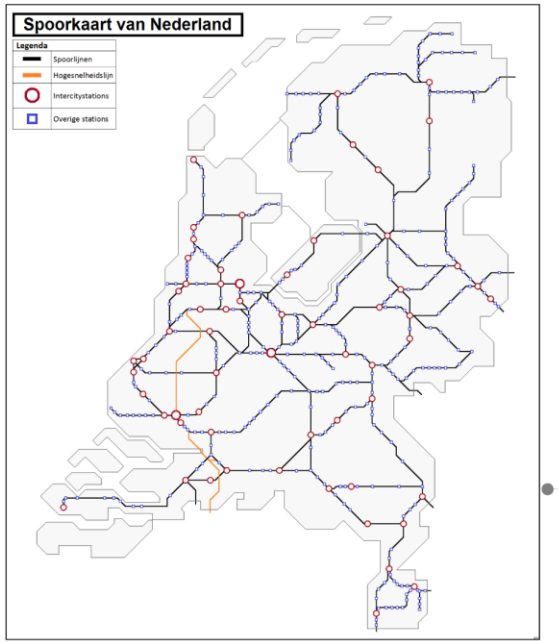
@Farah, Abdallah and Shahrour (2017)

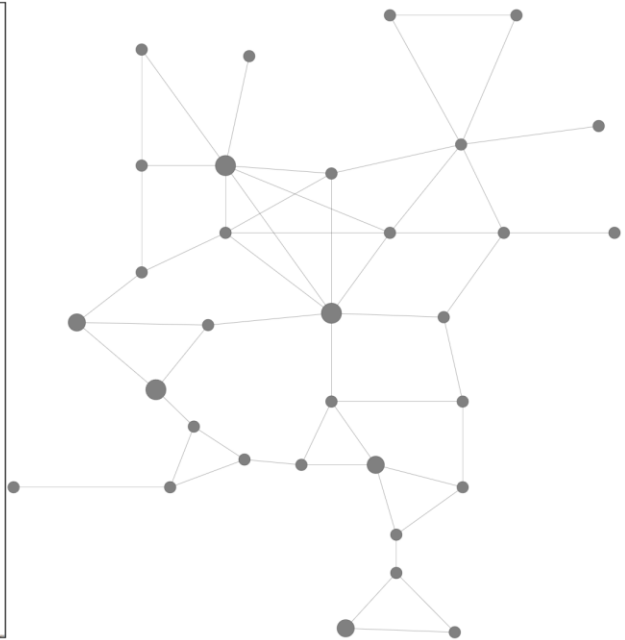
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Simplified railroad network in the Netherlands

Core Concepts:

Nodes, Edges



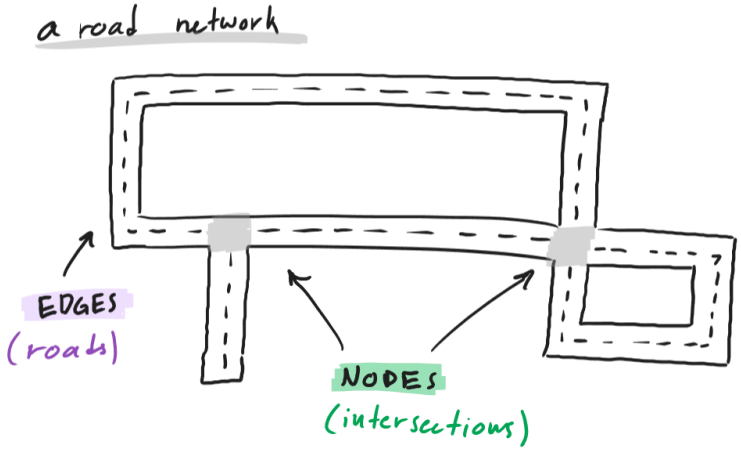


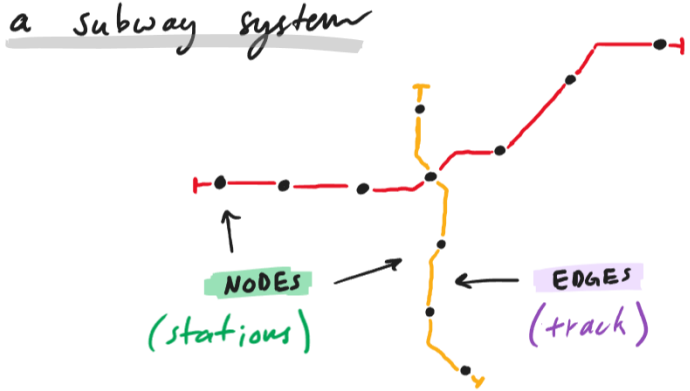
@Thenetworkpages

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Nodes & Edges

- **Nodes** (also called vertices) are the fundamental entities or "points" within a network
- **Edges** (also called arcs, or links) represent the connections or relationships between two nodes in a network
- **Topological rules** (e.g., connectivity) define how features are linked. This is what distinguishes a GIS network from a simple collection of lines in a drawing program. It's the **intelligence that enables analysis**





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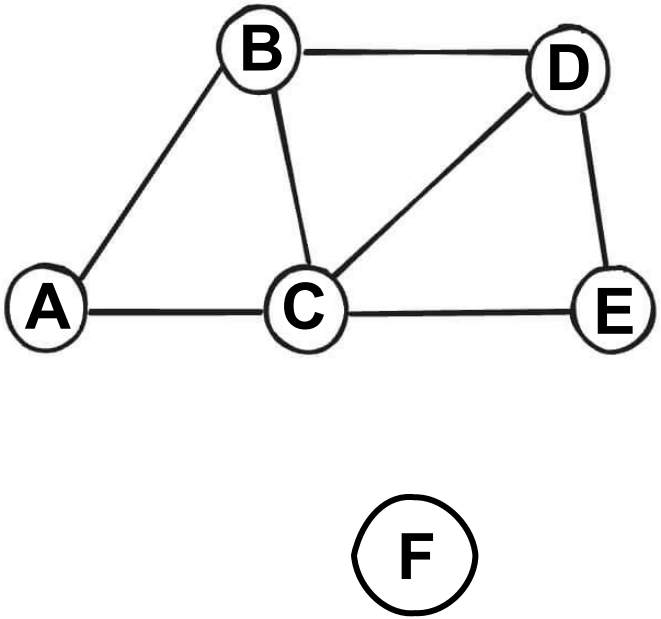
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Degree of Nodes

The **degree** of a node is the **number of edges connected** to it

A ➡️ 2
C ➡️ 4
F ➡️ 0

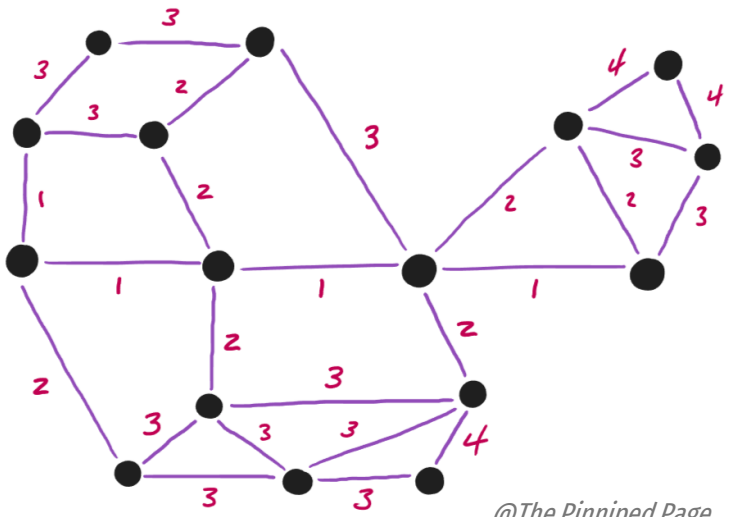
What is the degree of A, C, & F?



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Adding Richness to Networks – Attributes, Weights & Cost

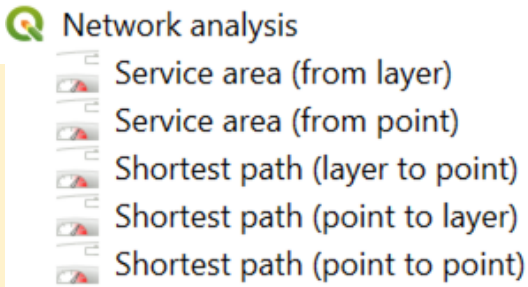
- A **weighted graph** is a graph in which each **edge is assigned a weight or cost**, i.e., a value
- Weight can be a distance, a travel time, an energy consumption, a difficulty, or risks, etc.
- Many **network analyses aim** to **find paths** or solutions that **minimize** this **accumulated cost** (e.g., the shortest path in terms of time or distance)



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Network analysis & QGIS

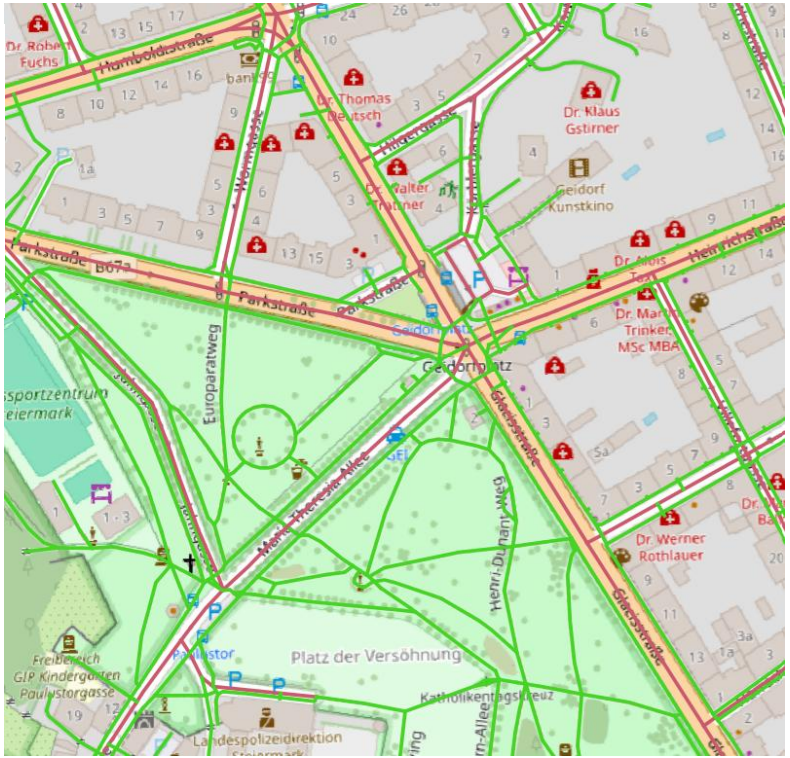


A network is made of:

- Nodes, connection points (intersections)
- Edges, paths between nodes (roads)

QGIS can compute:

- Shortest path
- Fastest path
- Service area



@QGIS documentation

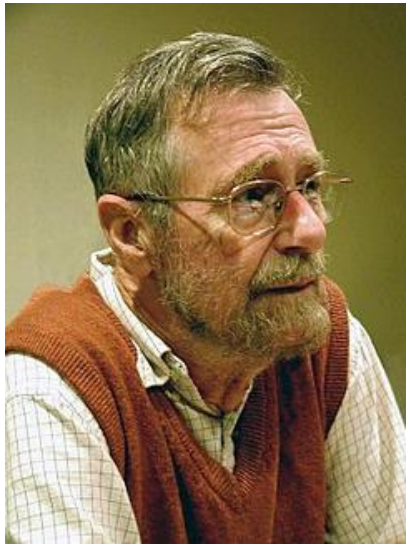
Shortest path

~ Questions related to the shortest path analysis ~

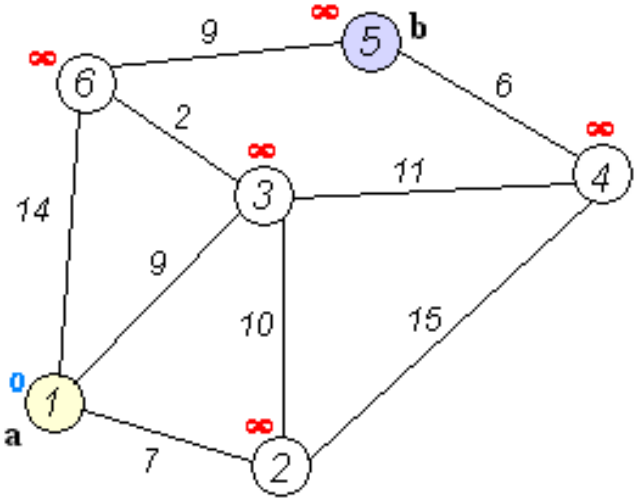
What is the fastest route from home to work?

How can city planners design road networks to minimize congestion and improve mobility?

How can utility companies find the shortest route for laying cables or pipes between two points?



Dijkstra's Algorithm

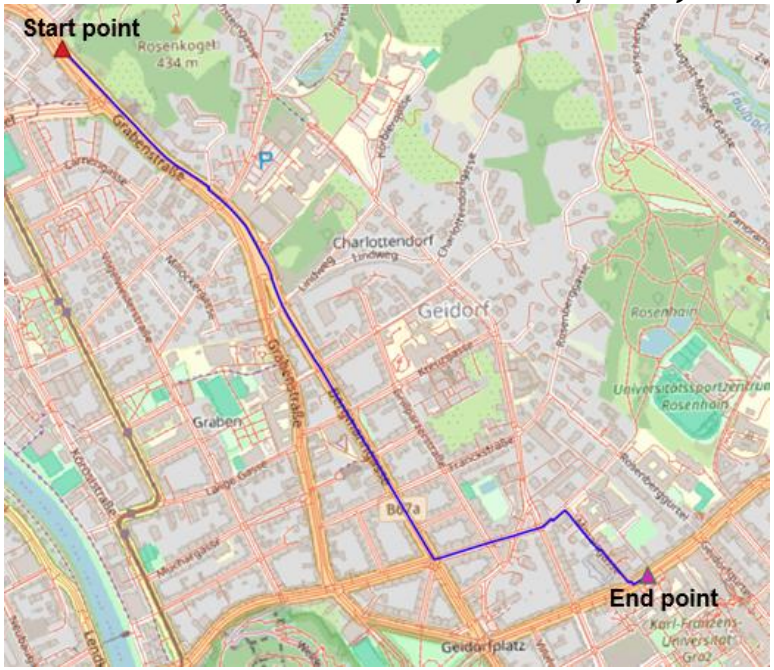


@Wikipedia

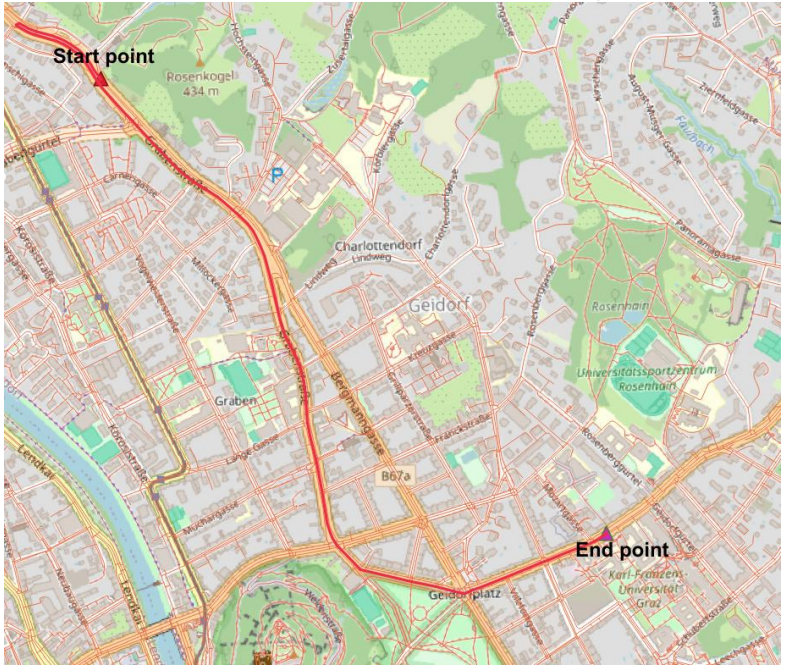
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Time & distance routes

What is the shortest path from start point to end point by foot ?



What is the fastest path from start point to end point by car ?



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Takeaway

Coordinate Systems	Network Analysis
“WHERE I AM”	“HOW I GET THERE”
Tells you exactly where things are on a map	Shows you the best way to get from one place to another

Why do both topics rely on spatial accuracy?

Accurate routing depends on correct spatial referencing!

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